



Ambasáid na hÉireann | Vítneam  
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## CONCEPT NOTE

### Vietnam Ireland Bilateral Education Exchange (VIBE)

2026

1. **Project Title:** Alzheimer's research in Vietnam and Ireland: Detection algorithms (**AVID**)

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#### 2. **Rationale -Why the proposed intervention is important and relevant**

Dementia affected 57 million individuals worldwide in 2021, primarily in low- and middle-income countries [1]. By 2049, Vietnam's elderly population is projected to reach 19.6 million, accounting for 18.1% of the population [2, 3]. Alzheimer's Disease (AD) is responsible for 60-70% of dementia cases [1], typically preceded by the Mild Cognitive Impairment stage (MCI) [4]. MCI affects more individuals over 60 in rural Vietnam than in urban areas, highlighting the critical need for early detection [5]. Currently, accurate diagnostic methods such as MRI and PET scans are costly and unevenly distributed, making them largely inaccessible in rural Vietnam [6, 7]. Therefore, there is a diagnosis gap in the millions of Vietnamese elderly people who cannot receive the right therapeutic interventions until late-stage dementia. Additionally, in Ireland, they also face an identical demographic which motivates this collaborative research project.

## Impact- What does the project aim to accomplish?

**Primary Objective.** The main goal of AVID is the development of an AI assistant for Alzheimer’s clinicians and researchers to deliver new levels of predictive capabilities. This tool will be designed to detect Alzheimer’s from a patient’s speech.

**Outcomes and Results.** The AVID collaboration seeks to develop the first validated, Vietnamese language voice dataset collected from AD, MCI, and healthy control participants. This dataset and resulting AI models will serve as a foundation for future speech biomarker studies in Vietnamese cognitive health. The Vietnamese team will gain hands-on capacity in advanced feature extraction, machine learning model development, and AI validation methodology through bilateral workshops and joint supervision with Dublin City University. Our Vietnamese and AI model will be developed to easily adapt, scale, and conduct further research in the northern and central Vietnamese clinical environment, offering a pathway to early screening across the diversity of Vietnamese healthcare settings. Another point worth considering is that this work establishes a replicable framework for voice-based dementia screening in low- and middle-income countries across Southeast Asia.

**Table 1: Deliverables.**

| Ref | Month | Description                                                                                                               | Type    |
|-----|-------|---------------------------------------------------------------------------------------------------------------------------|---------|
| D1  | 3     | Alzheimer’s and Data workshop in <b>HCMIU</b> . The output is a detailed project report.                                  | Report  |
| D2  | 6     | Development of first prototype (AVID v.1) of predictive tool.                                                             | AI Tool |
| D3  | 10    | AI Model Development. workshop in Dublin ( <b>DCU</b> ). The output is a detailed report focused on AI model development. | Report  |
| D4  | 12    | Development of final prototype (AVID v.2). Minor changes are expected until end of project.                               | AI Tool |
| D5  | 15    | Final Workshop ( <b>HCMUT</b> ). Main agenda topics are Summary of Results and Sustainability for the AVID project.       | Report  |

### 3. Approach- How will the objectives be achieved?

**Approach.** AVID will involve a number of steps (project development stages) which are interlinked with a series of mobilisation for the purpose of workshops and planning meetings. The project comprised 4 steps: Step 1. Acquisition of Patient speech data; Step 2. Data Preparation and Feature Extraction; Step 3. AI Model Development; and Step 4. Model Validation, Impact Assessment, and Interpretability.

**Project Duration.** The project will run for 15 months, starting on October-1-2026 and completing on December-31-2027.

**Mobilization of Resources, Researchers and Knowledge.**

The AVID project will mobilize resources through a structured bilateral exchange framework designed to strengthen institutional collaboration between Vietnam and Ireland, with a focus on capacity building in AI-driven prediction of Alzheimer's disease using voice data.

**Human resources:** The project will establish a joint Vietnam–Ireland research team comprising principal investigators, researchers, PhD candidates, and graduate students from the three participating institutions with that target of 50% female participation. Each partner institution will designate core academic staff to lead thematic activities, coordinate workshops, and supervise collaborative research outputs. Participating institutions will contribute academic research time, laboratory access, computational infrastructure, and supervisory engagement.

**Knowledge exchange and Network Mobilization** are operationalized through regular online technical meetings and 3 bilateral workshops:

- Workshop 1 (Vietnam, kick off meeting) – Theme: Alzheimer's and Data. This workshop will introduce current advances in Alzheimer's research and focus on voice data acquisition, clinical assessment alignment, ethical approvals, and harmonized data collection standards. The 2nd half of the workshop will focus on Feature Extraction from Voice Signals. This will address data preprocessing pipelines, signal segmentation, and acoustic and linguistic feature engineering. Existing technical approaches and preliminary analyses will be reviewed and standardized.
- Workshop 2 (Ireland) – Theme: AI Model Development. This workshop will focus on the development of machine learning models for predicting Alzheimer's disease using acoustic and linguistic features derived from voice data. It will be devised as a training workshop for Vietnamese researchers using resources at Dublin City University.
- Workshop 3 (Vietnam, final AVID meeting) – Theme 1: Model Validation and Interpretability. Theme 2: AVID Sustainability. The final workshop will integrate the analytical pipelines (from voice data preprocessing to predictive modeling) and address model validation, explainable AI, and potential clinical relevance. The 2nd half of the workshop will focus on sustainability for the AVID project, targeting new funding sources, eg. [8].

**Stakeholder Involvement in AVID.** In addition to the core academic partners, the project will engage clinical collaborators in Vietnam to support patient recruitment, voice data collection, and clinical interpretation of results. Postgraduate researchers will be involved in the workshops and research activities to strengthen training and capacity building. Also, potential engagement with digital health and AI stakeholders may be explored to support potential translational pathways for voice-based diagnostic technologies.

#### 4. What is the current status of the cooperation and the added value of the proposed partnership?

##### **Current Status.**

The collaboration was formed by Prof. Tho Quan Thanh during a visit to Dublin to attend the ACM Multimedia (October 2025) conference who used the opportunity for a series of meetings with Prof. Roantree. After an initial investigation of the project's feasibility, the group met again in Ho Chi Minh city when Prof. Roantree was presenting at the MIWAI conference in Ho Chi Minh City (November 2025). At this meeting, Dr. Ha Thi Thanh Huong agreed to lead the project. All 3 agreed to provide researchers to the project.

##### **Partner Complementary Expertise.**

The project lead, Prof. Ha Thi Thanh Huong, has a strong foundation in Alzheimer's research, including work on acoustic and language-based biomarkers for the Alzheimer's English dataset (DementiaBank) [15,16], and was won a number of awards for her gamified interventions on Dementia [15-17]. Prof. Tho contributes his expertise in natural language processing and adapts a multilingual AI model and its features for the Vietnamese setting [13,14]. Prof. Roantree has extensive experience developing AI models and has previously worked in dementia detection [9,10]. For the AVID project, the leads bring together a set of crucial yet complementary skills. Prof. Huong's will bring domain (Alzheimer's) expertise; Prof. Tho will bring the necessary skills to perform feature extraction on raw audio data which both prepares the data for machine learning but also anonymizes data so that it can be shared with the Irish partner. Prof Roantree's team will bring significant machine learning expertise. This delivers an added value which cannot be achieved by individual partners working in isolation. Our current team comprises the 3 leaders (1 female and two male); assisted by young researchers Phan Ngoc Minh Thu (F) and Vu Nguyen Minh Duc (M) from HCMIU; Vo Thi Nhu Quynh (F) from HCMUT; and Dongyun Nie (F) from DCU.

#### 5. Sustainability

Sustainability for AVID research will be a continuous effort during the lifetime of the project. The final workshop will contain a 1-day session on target funding sources and plans for the development of new funding proposals, where one possible option has already been identified as [8].

#### 6. What is the preliminary budget (€)?

|                                               |         |
|-----------------------------------------------|---------|
| <b>Project budget</b>                         | €86,651 |
| <i>Fund requested from Embassy of Ireland</i> | €50,800 |
| <i>Contribution from Vietnamese partner</i>   | €8,000  |
| <i>Contribution from Irish partner</i>        | €27,851 |
| <i>Funding from other sources</i>             | €0      |

#### 7. Have you ever submitted this project idea to any other funders?

No. This is an early stage collaboration and it is anticipated that VIBE funding will develop our idea to a level where we can apply for funding.

## Appendix I: References

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